

SUBJECTIVE QUESTIONS

1. A segment of a charged wire of length l , charge density λ_2 and an infinitely long charged wire of charge density λ_1 lie in a plane at right angles to each other. The separation between the wires is r_0 . Determine the force of interaction between the wires.

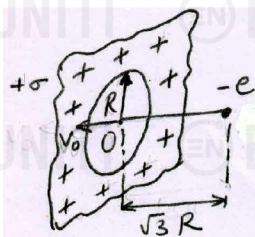
2. A particle of mass $m = 0.1$ kg and having positive charge $q = 75 \mu\text{C}$ is suspended from a point by a thread of length $l = 10$ cm. In the space, a uniform horizontal electric field $E = 10^4 \text{ NC}^{-1}$ exists. The particle is drawn aside so that the thread becomes vertical and then it is projected horizontally with velocity v such that the particle starts to move along a circle with the same constant speed v . Calculate the radius of the circle and speed v (Take $g = 10 \text{ ms}^{-2}$).

3. Three particles, each of mass 1 g and carrying a charge q , are suspended from a common point by insulated massless strings, each 100 cm long. If the particles are in equilibrium and are located at the corners of an equilateral triangle of side length 3 cm, calculate the charge q on each particle. (Take $g = 10 \text{ m/s}^2$).

4. Two balls of charges q_1 and q_2 initially have same speed u in the same direction. After a uniform electric field is applied for some time, the direction of velocity of the first ball changes by 60° and the magnitude is reduced to half i.e., $v_1 = u/2$. The direction of velocity of the second ball changes by 90° . There is no gravity.

(a) What is the final speed v_2 of the second ball?
(b) If the two balls have same mass, what is the ratio of their charges q_1/q_2 ?

5. An infinite dielectric sheet having charge density σ has a hole of radius R in it. An electron of charge $-e$ and mass m is released on the axis of the hole at a distance $\sqrt{3}R$ from the centre. What will be the velocity with which it crosses the plane of sheet?



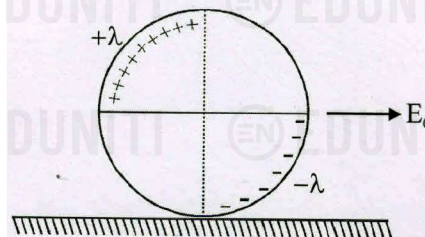
6. Two small balls having the same mass and charge are located on the same vertical at heights h_1 and h_2 . They are thrown in the same direction along the horizontal with same velocity v . The first ball hits the ground at a horizontal distance L from the initial vertical. At what height will the second ball be at this instant? The air drag and the charges induced should be neglected.

7. Two fixed charges $-2Q$ and $+Q$ are located at the points with coordinates $(-3a, 0)$ and $(+3a, 0)$ respectively in the x - y plane.

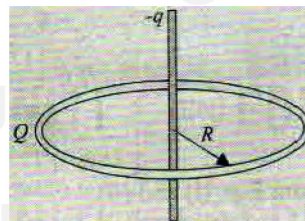
(a) Show that all points in the x - y plane where the electric potential due to the two charges is zero, lie on a circle. Find its radius and the location of its centre.
(b) If a particle of charge $+q$ starts from rest at the centre of the circle, show by a short quantitative argument that the particle eventually crosses the circle. Find its speed when it does so.

8. Figure shows three thin concentric spherical shells A, B, and C with initial charges on them as $3Q$, $2Q$, and $-Q$, respectively. The shells A and C are now connected by a wire such that it does not touch B and shell B is earthed. Determine the final charges q_A , q_B , and q_C .

9. A nonconducting ring of mass m and radius R is charged with charge per unit length $+\lambda$ on one fourth ring and $-\lambda$ on opposite one fourth ring as shown in figure. It is placed on a rough nonconducting horizontal plane. At time $t = 0$, a uniform horizontal electric field E_0 is switched on and the ring starts rolling without sliding. Determine the acceleration of the ring axis and friction force acting on the ring, when it starts moving.



10. In a free space, a thin rod carrying uniformly distributed negative charge $-q$ is placed symmetrically along the axis of a thin ring of radius R carrying uniformly distributed charge Q . The mass of the rod is m and length is $\ell = 2R$. The ring is fixed and the rod is free to move. The rod is displaced slightly along the axis of the ring and then released. Find the time period T of small oscillations of the rod.

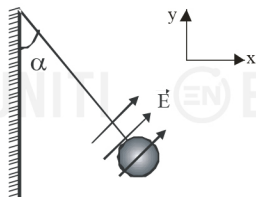


11. A non-conducting ring of mass m and radius r is lying at rest in the vertical xy plane on a smooth non-conducting horizontal xz plane. Charges $+q$ and $-q$ are distributed uniformly on the ring on the two sides of the vertical diameter of the ring. A constant and uniform electric field $\vec{E}$ is set up along the x -direction. The ring is given a small rotation about the vertical diameter of the ring and released. Find the time period of oscillation of the ring.

MORE THAN ONE CORRECT QUESTIONS

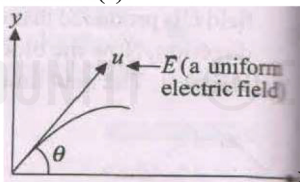
12. A charged cork of mass m suspended by a light string is placed in uniform electric field of strength $\vec{E} = (\hat{i} + \hat{j}) \times 10^5 \text{ NC}^{-1}$ as shown in the figure. If in equilibrium

position, the tension in the string is $\frac{2mg}{1+\sqrt{3}}$, then the angle α with the vertical is



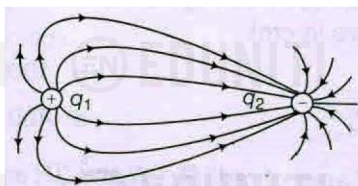
- (a) 60° (b) 30°
(c) 45° (d) 18°

13. A particle of mass m and charge q is projected from ground at an angle θ to horizontal in xy plane as shown in fig. Mark the correct statement(s).



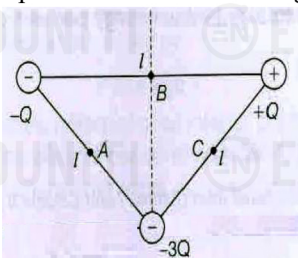
- (a) The path of motion of the particle may be a parabola.
(b) The path of motion of the particle may be a straight line.
(c) Time of flight of the particle is $2u \sin \theta / g$
(d) Time of flight is less than $2u \sin \theta / g$

14. Two charges form a field pattern as shown. The relation between q_1 and q_2 satisfy,



- (a) $|q_1| = |q_2|$ (b) $|q_1| < |q_2|$
(c) $|q_1| > |q_2|$ (d) A negative test charge can be placed on left side of positive charge such that it experiences zero force

15. Three charges are at the corners of an equilateral triangle. A , B and C are mid-points of the sides of triangle.



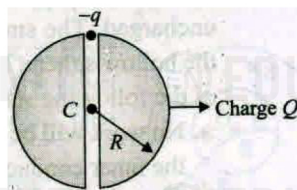
Mark the correct option(s).

(a) $V_A = \frac{Q}{\pi \epsilon_0 \ell} \left(\frac{\sqrt{3}}{6} - 2 \right)$ (b) $V_B = \frac{\sqrt{3} Q}{2 \pi \epsilon_0 \ell}$

(c) $V_C = -\frac{Q}{\pi \epsilon_0 \ell} \left(1 + \frac{\sqrt{3}}{6} \right)$

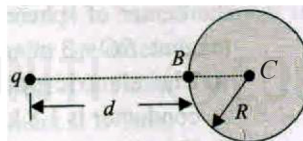
- (d) Total potential energy of system of charges is greater than zero

16. In a uniformly charged dielectric sphere, a very thin tunnel has been made along the diameter as shown in fig. A charged particle $-q$ having mass m is released from rest at one end of the tunnel. For the situation described, mark the correct statement(s). (Neglect gravity.)



- (a) Charged particle will perform SHM about center of the sphere as mean position.
(b) Time period of the particle is $2\pi \sqrt{4\pi \epsilon_0 m R^3 / qQ}$
(c) Particle will perform oscillation but not SHM
(d) Speed of the particle while crossing the mean position is $\sqrt{qQ / 4\pi \epsilon_0 m R}$

17. For the situation shown in the figure, select the correct statement (s).

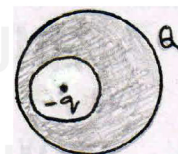


- (a) Potential of the conductor is $q / 4\pi \epsilon_0 (d + R)$
(b) Potential of the conductor is $q / 4\pi \epsilon_0 d$
(c) Potential of the conductor cannot be determined as nature of distribution of induced charges is not known.
(d) Potential at point B due to induced charges is $-qR / 4\pi \epsilon_0 (d + R)d$

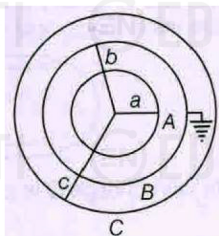
18. A solid conductor has a cavity.

Conductor carries charge $+Q$ and a point charge $-q$ lies in the centre of cavity. Then,

- (a) charge $(Q - q)$ is present on the outer surface of conductor
(b) charge $-q$ is present on the outer surface of conductor
(c) net field inside the conductor is zero
(d) net field inside the cavity is zero



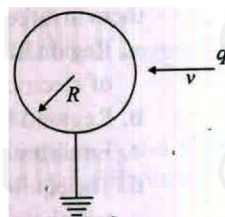
- 19 The given figure shows three concentric thin spherical shells A , B and C of radii a , b and c . The shells A and C are given charges q and $-q$, respectively, and shell B is earthed. Then,



- (a) the charge q will be there at outer surface A
- (b) the charge $-bq/c$ will be there at inner surface of C
- (c) the charge $-q$ will be at inner surface of B
- (d) the charge $q(1 - b/c)$ will be there at outer surface of C

20. A charged particle having a positive charge q approaches a grounded metallic sphere of radius R with a constant small speed v as shown in fig. In this situation,

- (a) a current flows into the ground
- (b) a current flows out of the ground into the sphere
- (c) the magnitude of the current increases
- (d) the magnitude of the current decreases



21. Electric field inside a charged spherical ball depends only on distance from the centre, as $E(r) = ar$, where a is a constant. If $\rho(r)$ is the volume charge density at distance r and $d\phi$ is the flux through a thin spherical layer of thickness dr and radius r , then

- (a) $d\phi = \rho(r)4\pi r^2 dr / \epsilon_0$
- (b) $d\phi = 12\pi ar^2 dr$
- (c) $\rho(r) = 3a\epsilon_0$
- (d) $\rho(r) = 3a\epsilon_0 / r$

22. Imagine a short dipole at the center of an imaginary spherical surface. If the magnitude of the electric field at a certain point on the surface of the sphere is 10 NC^{-1} , then which of the following cannot be the magnitude of the electric field anywhere on the surface of the sphere?

- (a) 4 NC^{-1}
- (b) 8 NC^{-1}
- (c) 16 NC^{-1}
- (d) 32 NC^{-1}

PARAGRAPH TYPE QUESTIONS

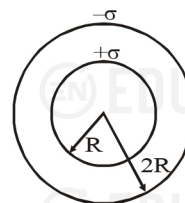
Paragraph for Question Nos. 23 to 24

An insulated ring having a charge $q_1 = 2 \times 10^{-5} \text{ C}$ uniformly distributed over it has radius 4m . Another particle having charge $q_2 = 4 \times 10^{-4} \text{ C}$ is released from rest along its axis at distance $x = 3\text{m}$ from its centre. Mass of both ring and the particle is 1kg each. Neglect gravitational effects. Both ring and particle are free to move.

- 23. The maximum speed of the particle will be
 - (a) 5.37 m/s
 - (b) 3.79 m/s
 - (c) 2.68 m/s
 - (d) 1.89 m/s
- 24. If ring is not free to move, then maximum speed of particle will be
 - (a) 5.37 m/s
 - (b) 3.79 m/s
 - (c) 2.68 m/s
 - (d) 1.89 m/s

Paragraph for Question Nos. 25 to 27

Two thin concentric conducting spherical shells have surface charge densities $+\sigma$ and $-\sigma$ as shown in the figure. Their radii are R and $2R$ respectively. Now they are connected by a thin conducting wire.



25. Potential on the inner shell will be

- (a) $-\frac{3\sigma R}{2\epsilon_0}$
- (b) $-\frac{2\sigma R}{2\epsilon_0}$
- (c) $-\frac{\sigma R}{2\epsilon_0}$
- (d) zero

26. If electric field at a distance $r (> 2R)$ was E_1 before connecting the two shells and E_2 after connecting the two shells, then $|E_2/E_1|$ is

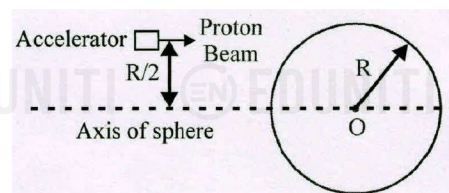
- (a) zero
- (b) 1
- (c) 2
- (d) $1/2$

27. In electric field at a distance $r = 3R/2$ was E_1 before connecting the two shells and E_2 after connecting the two shells, then $|E_2/E_1|$ is

- (a) zero
- (b) 1
- (c) $9/8$
- (d) $8/9$

Paragraph for Question Nos. 28 to 29

An accelerator produces a narrow beam of protons, each having an initial speed of v_0 . The beam is directed towards an initially uncharged metal sphere of radius R and centered at point O . The sphere is fixed and is at a large distance from the accelerator. The initial path of the beam is parallel to the axis of the sphere at a distance of $R/2$ from the axis, as shown in the figure.



The protons in the beam that collide with the sphere will cause it to become charged. The mass of the proton is m_p and the charge on it is e . The potential of the sphere increases with time and eventually reaches a constant value due to the proton beam getting deflected upwards without touching the sphere. Answer the questions given below after the sphere has attained a constant potential.

- 28. The minimum speed v of a proton along its trajectory path is
 - (a) v_0
 - (b) $v_0/2$
 - (c) $2v_0$
 - (d) None of these

29. The electric potential of the sphere is

- (a) $\frac{3m_p v_0^2}{8e}$ (b) $\frac{3m_p v_0^2}{4e}$
(c) $\frac{3m_p v_0^2}{2e}$ (d) None of these

Paragraph for Question Nos. 30 to 31

A system consists of a ball of radius R carrying a spherically symmetrical charge q and the surrounding space filled with a charge of volume density $\rho = \alpha/r$, where α is constant and r is the distance from the centre of the ball.

30. Magnitude of the electric field as a function of r for $r > R$ is

- (a) $E = \frac{\rho}{\epsilon_0}$
(b) $E = \frac{1}{4\pi\epsilon_0} \cdot \frac{(4/3\pi r^3) \cdot \rho}{R^2}$
(c) $E = \left(\frac{q}{4\pi\epsilon_0} - \frac{\alpha R^2}{2\epsilon_0} \right) \cdot \frac{1}{r^2} + \frac{\alpha}{2\epsilon_0}$
(d) $E = \frac{\alpha R^2}{2\epsilon_0 r^2} - \frac{\alpha}{2\epsilon_0}$

31. Electric field in the region outside the ball is independent of distance r when charge on ball is

- (a) $q = \frac{4}{3}\pi\alpha R^3$ (b) $q = \frac{4}{3}\alpha R^2$
(c) $q = 4\pi\alpha R^2$ (d) $q = 2\pi\alpha R^2$

Paragraph for Question Nos. 32 to 34

Consider a volume charge with density $\rho = \rho_0 e^{-\alpha r^3}$, where ρ_0 and α are constants and r is distance from the centre of the charge density.

32. Electric field strength as a function of r is

- (a) $\frac{4\rho_0}{3\alpha\epsilon_0 r^2} (1 - e^{-\alpha r^3})$ (b) $\frac{\rho_0}{3\alpha\epsilon_0 r^2} (e^{-\alpha r^3})$
(c) $\frac{\rho_0}{3\alpha\epsilon_0 r^2} (1 - e^{-\alpha r^3})$ (d) $\frac{\rho_0}{3\alpha\epsilon_0 r^2} \cdot 2e^{-\alpha r^3}$

33. If $\alpha r^3 \gg 1$, then the field at a distance r from centre is

- (a) $\frac{\rho_0 r}{3\epsilon_0}$ (b) $\frac{4\rho_0 \epsilon}{3\epsilon_0}$
(c) 0 (d) $\frac{\rho_0 e^{-\alpha r^3}}{\epsilon_0}$

34. If $\alpha r^3 \gg 1$, then the field at a distance r from the centre is

- (a) $\frac{\rho_0}{3\alpha\epsilon_0 r^2}$ (b) $\frac{\rho_0}{\alpha\epsilon_0 r^2}$
(c) $\frac{\rho_0 e^{-\alpha r^3}}{3\alpha\epsilon_0 r^2}$ (d) $\frac{4\rho_0 e^{-\alpha r^3}}{3\alpha\epsilon_0 r^2}$

Paragraph for Question Nos. 35 to 37

Two equal point charges of $8\mu\text{C}$ are fixed at points $(2\text{ m}, 0)$ and $(-2\text{ m}, 0)$. Another charge q with a mass of 91 mg is released at point $(0, 0.1\text{ m})$ at $t = 0$. It is observed that q oscillates in simple harmonic motion about the origin. At $t = 0$, force experienced by q is $9 \times 10^{-3}\text{ N}$.

35. Charge q is

- (a) $-8\mu\text{C}$ (b) $-6.5\mu\text{C}$
(c) $-5\mu\text{C}$ (d) $+6.5\mu\text{C}$

36. Amplitude of simple harmonic motion is

- (a) 10 cm (b) 20 cm
(c) 30 cm (d) 40 cm

37. Frequency of oscillation is

- (a) 8 Hz (b) 10 Hz
(c) 5 Hz (d) 2 Hz

Paragraph for Question Nos. 38 to 40

A non conducting massless rod of length L has two small balls of mass m and charge Q attached to its ends. The rod can rotate in the horizontal plane about a vertical axis passing through the line joining the two balls at a distance $L/4$ from one of its ends.

38. Initially, the rod is in unstable equilibrium in a horizontal uniform electric field E . When it is gently displaced from this position, determine the maximum velocity attained by the ball that is closer to the axis in the subsequent motion.

- (a) $\sqrt{\frac{2QEL}{m}}$ (b) $\sqrt{\frac{2QEL}{5m}}$
(c) $\sqrt{\frac{QEL}{5m}}$ (d) $\sqrt{\frac{4QEL}{5m}}$

39. In what position is the rod to be set so that if displaced a little from that position, it begins a harmonic oscillation about the axis A ?

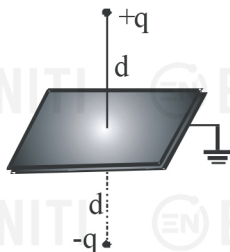
- (a)  (b) 
(c)  (d) 

40. What is the time period of the SHM for small angular displacement from equilibrium position?

- (a) $2\pi\sqrt{\frac{mL}{QE}}$ (b) $2\pi\sqrt{\frac{2mL}{3QE}}$
(c) $2\pi\sqrt{\frac{5mL}{8QE}}$ (d) $2\pi\sqrt{\frac{5mL}{4QE}}$

Paragraph for Question Nos. 41 to 42

The method of electrical images is used to solve the electrostatic problems, where charge distribution is not specified completely. The method consists of replacement of given charge distribution by a simplified charge distribution provided that the original boundary conditions are still satisfied. For example, consider a system containing a point charge $+q$ placed at a distance d from an infinite earthed conducting plane. Here, the boundary conditions are



(i) Potential is zero at infinity

(ii) Potential is zero at each point on the conducting plane

Now, if we replace the conducting plane by a point charge $-q$ placed at a distance d opposite to conducting plane, the charge $-q$ is called the electrical image of charge $+q$.

Now, the system consists of two charges $+q$ and $-q$ at separation $2d$. It can be seen that the original boundary conditions still remain the same.

41. The force between point charge $+q$ and earthed conducting plane is

(a) $\frac{1}{4\pi\epsilon_0} \frac{q^2}{d^2}$ (repulsive) (b) $\frac{1}{4\pi\epsilon_0} \frac{q^2}{d^2}$ (attractive)

(c) $\frac{1}{4\pi\epsilon_0} \frac{q^2}{4d^2}$ (repulsive) (d) $\frac{1}{4\pi\epsilon_0} \frac{q^2}{4d^2}$ (attractive)

42. The potential energy of system of charge $+q$ placed at a distance d from the earthed conducting plane is

(a) $\frac{1}{4\pi\epsilon_0} \frac{q^2}{d^2}$ (b) $-\frac{1}{4\pi\epsilon_0} \frac{q^2}{2d^2}$

(c) $\frac{1}{4\pi\epsilon_0} \frac{q^2}{2d}$ (d) $-\frac{1}{4\pi\epsilon_0} \frac{q^2}{4d}$